

Q1 2021 (01 Sep Shift 2)

A steel rod with $y = 2.0 \times 10^{11} \text{ Nm}^{-2}$ and $\alpha = 10^{-5} \text{ }^\circ\text{C}^{-1}$ of length 4 m and area of cross-section 10 cm^2 is heated from 0°C to 400°C without being allowed to extend. The tension produced in the rod is $x \times 10^5 \text{ N}$ where the value of x is

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Q2 2021 (01 Sep Shift 2)

Due to cold weather a 1 m water pipe of cross-sectional area 1 cm^2 is filled with ice at -10°C . Resistive heating is used to melt the ice. Current of 0.5 A is passed through $4 \text{ k}\Omega$ resistance. Assuming that all the heat produced is used for melting, what is the minimum time required ?

(Given latent heat of fusion for water/ice $= 3.33 \times 10^5 \text{ J kg}^{-1}$, specific heat of ice $= 2 \times 10^3 \text{ J kg}^{-1}$ and density of ice $= 10^3 \text{ kg/m}^3$)

(1) 0.353 s

(2) 35.3 s

(3) 3.53 s

(4) 70.6 s

Q3 2021 (31 Aug Shift 2)

Two thin metallic spherical shells of radii r_1 and r_2 ($r_1 < r_2$) are placed with their centres coinciding. A material of thermal conductivity K is filled in the space between the shells. The inner shell is maintained at temperature θ_1 and the outer shell at temperature θ_2 ($\theta_1 < \theta_2$). The rate at which heat flows radially through the material is :-

(1) $\frac{4\pi K r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$

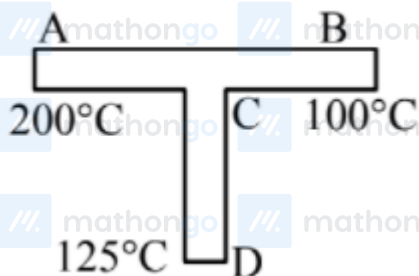
(2) $\frac{\pi r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$

(3) $\frac{K(\theta_2 - \theta_1)}{r_2 - r_1}$

(4) $\frac{K(\theta_2 - \theta_1)(r_2 - r_1)}{4\pi r_1 r_2}$

Q4 2021 (27 Aug Shift 1)

A rod CD of thermal resistance 10.0 KW^{-1} is joined at the middle of an identical rod AB as shown in figure, The end A, B and D are maintained at 200°C , 100°C and 125°C respectively. The heat current in CD is P watt. The value of P is



Q5 2021 (27 Aug Shift 1)

An ideal gas is expanding such that $PT^3 = \text{constant}$. The coefficient of volume expansion of the gas is :

- (1) $\frac{1}{T}$
- (2) $\frac{2}{T}$
- (3) $\frac{4}{T}$
- (4) $\frac{3}{T}$

Q6 2021 (26 Aug Shift 2)

The temperature of equal masses of three different liquids x, y and z are 10°C , 20°C and 30°C respectively. The temperature of mixture when x is mixed with y is 16°C and that when y is mixed with z is 26°C . The temperature of mixture when x and z are mixed will be :

- (1) 28.32°C
- (2) 25.62°C
- (3) 23.84°C
- (4) 20.28°C

Answer Key

Q1 (8)

Q2 (2)

Q3 (1)

Q4 (2)

Q5 (3)

Q6 (3)

Q1 (8)

Thermal force $F = A\gamma \propto \Delta T$

$$F = (10 \times 10^4) (2 \times 10^{11}) (10^{-5}) (400)$$

$$F = 8 \times 10^5 \text{ N}$$

$$\Rightarrow x = 8$$

Q2 (2)

mass of ice $m = \rho A \ell = 10^3 \times 10^{-4} \times 1 = 10^{-1} \text{ kg}$

Energy required to melt the ice

$$Q = ms \Delta T + mL$$

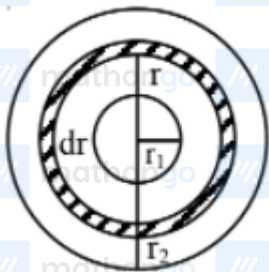
$$= 10^{-1} (2 \times 10^3 \times 10 + 3.33 \times 10^5) = 3.53 \times 10^4 \text{ J}$$

$$Q = i^2 R T \Rightarrow 3.53 \times 10^4 = \left(\frac{1}{2}\right)^2 (4 \times 10^3) (t)$$

$$\text{Time} = 35.3 \text{ sec}$$

Option (2)

Q3 (1)

Thermal resistance of spherical sheet of thickness dr and radius r is

$$dR = \frac{dr}{K(4\pi r^2)}$$

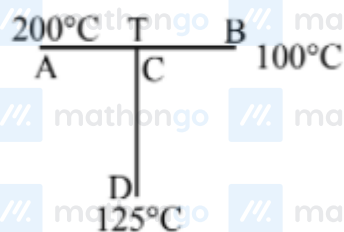
$$R = \int_{r_1}^{r_2} \frac{dr}{K(4\pi r^2)}$$

$$R = \frac{1}{4\pi K} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = \frac{1}{4\pi K} \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

$$\text{Thermal current (i)} = \frac{\theta_2 - \theta_1}{R}$$

$$i = \frac{4\pi K r_1 r_2}{r_2 - r_1} (\theta_2 - \theta_1)$$

Q4 (2)



Rods are identical so

$$R_{AB} = R_{CD} = 10 \text{Kw}^{-1}$$

C is mid-point of AB, so $R_{AC} = R_{CB} = 5 \text{Kw}^{-1}$ at point C

$$\frac{200 - T}{5} = \frac{T - 125}{10} + \frac{T - 100}{5}$$

$$2(200 - T) = T - 125 + 2(T - 100)$$

$$400 - 2T = T - 125 + 2T - 200$$

$$T = \frac{725}{5} = 145^\circ\text{C}$$

$$I_h = \frac{145 - 125}{10} \text{w} = \frac{20}{10} \text{w}$$

$$I_h = 2\text{w}$$

Q5 (3)

$$PV^3 = \text{constant}$$

$$\left(\frac{nRT}{V}\right) T^3 = \text{constant}$$

$$T^4 V^{-1} = \text{constant}$$

$$T^4 = kV$$

$$\Rightarrow 4 \frac{\Delta T}{T} = \frac{\Delta V}{V} \dots \dots \dots (1)$$

$$\Delta V = V \gamma \Delta T \dots \dots \dots (2)$$

comparing (1) and (2) we get

$$\gamma = \frac{4}{T}$$

Q6 (3)

X

$$m_1 = m$$

$$T_1 = 10^\circ\text{C}$$

*s*₁

Y

$$m_2 = m$$

$$T_2 = 20^\circ\text{C}$$

*s*₂

Z

$$m_3 = m$$

$$T_3 = 30^\circ\text{C}$$

*s*₃

when *x* & *y* are mixed, $T_f = 16^\circ\text{C}$

$$m_1 s_1 T + m_2 s_2 T_2 = (m_1 s_1 + m_2 s_2) T_f$$

$$s_1 \times 10 + s_2 \times 20 = (s_1 + s_2) \times 16$$

$$s_1 = \frac{2}{3} s_2$$

... (i)

when *y* & *z* are mixed, $T_f = 26^\circ\text{C}$

$$m_2 s_2 T + m_3 s_3 T_3 = (m_2 s_2 + m_3 s_3) T_f$$

$$s_2 \times 20 + s_3 \times 30 = (s_2 + s_3) \times 26$$

$$s_3 = \frac{3}{2} s_2$$

when *x* & *z* are mixed

$$m_1 s_1 T_1 + m_3 s_3 T_3 = (m_1 s_1 + m_3 s_3) T_f$$

$$\frac{2}{3} s_2 \times 10 + \frac{3}{2} s_2 \times 30 = \left(\frac{2}{3} s_2 + \frac{3}{2} s_2 \right) T_f$$

$$T_f = 23.84^\circ\text{C}$$

Ans (3)